How Ricardian Are We?

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The views expressed herein are those of the authors and do not necessarily represent the views of the Federal Reserve Bank of Kansas City or the Federal Reserve System.

Introduction

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  - Micro studies estimate generic MPC, but we need the *MPC out of Debt-Financed Transfers* (missing intercept problem)

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   - Demand shocks cause OLS bias towards Ricardian equivalence
   - Barnichon-Mesters: structural macro equations can be estimated using macro shocks as IVs!
   - Consumption function has many endogenous regressions, so we assemble a large number of well-identified shocks from the literature

Our Contributions (cont.)

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   - Households only internalize 1/5 of future taxes
   - GE model: government borrowing crowds out capital

Theoretical Framework

"The debts of a nation are debts due from the right hand to the left."

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"The people who pay the taxes never so estimate them, and therefore do not manage their private affairs accordingly."

— David Ricardo (1820)

Behavioral Consumption Function Ingredients

- Government budget constraint:

$$B_{t-1} + G_t = T_t + Q_t B_t$$

- Household budget constraint:

$$B_{t-1} + R_t^K K_{t-1} + Y_t^N = C_t + T_t + Q_t B_t + K_t$$

- Household Euler Equation

$$Q_t = \beta \tilde{\mathbb{E}}_t \left[\frac{u'(C_{t+1})}{u'(C_t)} \right] + Z_t^d$$

Notes: $\tilde{\mathbb{E}}_t$ is a (possibly) non-rational expectation operator, Z_t^d an exogenous wedge

Behavioral Consumption Function Ingredients (Linearized)

- Government budget constraint:

$$b_{t-1} = \tau_t - g_t + \overline{B} q_t + \beta b_t$$

- Household budget constraint:

$$n_{t-1} + y_t = c_t + \tau_t + q_t \overline{B} + \beta n_t$$

- Household Euler Equation

$$q_t = \beta \tilde{\mathbb{E}}_t [\gamma (c_t - c_{t+1})] + z_t^d$$

Notes: n_t is household financial net worth, steady state returns are β^{-1}, normalize $\overline{C} = 1$

Behavioral Consumption Function: Expectations

- What is $\tilde{\mathbb{E}}_t$?
  - Distorts the rational expectation $\mathbb{E}_t$; relatively general
  - Can apply to all time series or subset
  - Naive or sophisticated (i.e. without or with L.I.E.)

- v_t and $\tilde{v}_t$ denote present values, rational and perceived, e.g.

$$v_t^\tau = \tau_t + \beta \mathbb{E}_t[v_{t+1}^\tau] \qquad \tilde{v}_t^\tau = \tau_t + \beta \tilde{\mathbb{E}}_t[v_{t+1}^\tau]$$

- Assumption: perceived P.V. of future taxes is proportional to true P.V.

$$\tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^\tau] = \theta \mathbb{E}_t[v_{t+1}^\tau]$$

θ is the *behavioral attenuation* (key parameter)

The Behavioral Consumption Function

Proposition

If expectations satisfy $\tilde{\mathbb{E}}_t[\tilde{v}^{\tau}_{t+1}] = \theta \mathbb{E}_t[v^{\tau}_{t+1}]$ *then consumption is given by*

$$c_t = (1-\beta)\left(n_{t-1} - \theta b_{t-1} + \tilde{v}^y_t - (1-\theta)\tau_t - \theta v^g_t + \theta \overline{B} v^q_t\right) + \left(\frac{1}{\gamma} - (1-\beta)\overline{B}\right)\tilde{v}^q_t + \zeta_t$$

Notes: ζ_t exogenous demand shock (determined by intertemporal wedges)

Empirical Strategy

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- ... but to tell if people are Ricardian, you need the *MPC out of Debt-Financed Transfers* (MPC-DFT)
- Cannot be estimated in (existing) cross-sectional studies because of the *missing intercept problem*
- So we will estimate the consumption function with time series data

Empirical Strategy: Instrumental Variables

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- Solution: use macro shocks from the literature as IVs (Barnichon-Mesters)
- This plan brings several challenges: develop **B-HIVE** to resolve

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  - Especially problematic here: long horizons in v_t^y etc.

The Bayesian Hybrid Instrumental Variable Estimator

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B-HIVE: State-Space Components

1. A structural equation:

$$c_t = \phi_0 + \phi_n \, n_{t-1} + \phi_b \, b_{t-1} + \phi_\tau \, \tau_t + \sum_{j \in \{y,g,q\}} \tilde{\phi}_j \, v_t^j + \zeta_t$$

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2. A VAR structure for X_t (RHS variables + other observables)

$$X_t = \mu_X + A_1 X_{t-1} + \cdots + A_p X_{t-p} + G\varepsilon_t \quad , \quad \varepsilon_t \sim N(0, I)$$

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3. A measurement equation relating instruments w_t to structural shocks ε_t

$$w_t = \mu_w + M_X X_{t-1} + M\varepsilon_t + \eta_t \qquad \eta_t \sim N(0, \Sigma_\eta)$$

Coefficients can be time-dependent.

Application

Data

- National accounts, everything at household level
  - y_t is personal income (less transfers), τ_t is personal taxes (net of transfers), g_t is expenditures net of residual taxes, n_t is household net worth
  - Normalize everything relative to a nominal GDP trend
- Many many shocks (next slide)
- Baseline: let $\tilde{v}_t^y$, $\tilde{v}_t^q$ be proportional to rational expectations
  - Later: augment with survey forecasts

Classification: External, HFI, Narrative, SVAR

Monetary Policy Shocks: Jarociński and Karadi (2020), Miranda-Agrippino and Ricco (2021), Bauer and Swanson (2023), Swanson (2024), Aruoba and Drechsel (2024), Drechsel (2024)

Government Spending Shocks: Fisher and Peters (2010), Ramey (2016), Romer and Romer (2016), Fieldhouse et al. (2018), Fieldhouse and Mertens (2023)

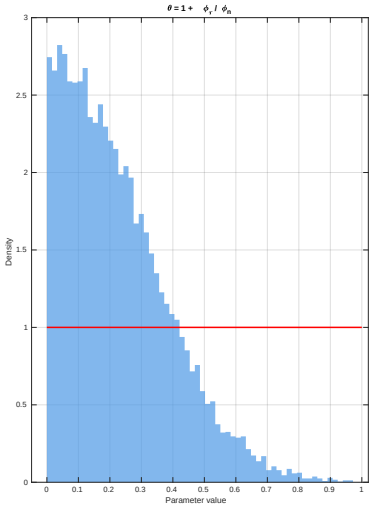
Tax/Borrowing Shocks: Leeper et al. (2012), Phillot (2025), Mertens and Ravn (2012), Lieb et al. (2024)

Technology Shocks: Fernald (2014), Miranda-Agrippino et al. (2025)

Oil Shocks: Kilian (2008), Känzig (2021), Baumeister and Hamilton (2019)

Other Shocks: Kim et al. (2025), Piffer and Podstawski (2018), Chahrour and Jurado (2022), Adams and Barrett (2024)

Posterior Distributions: Not Very Ricardian!



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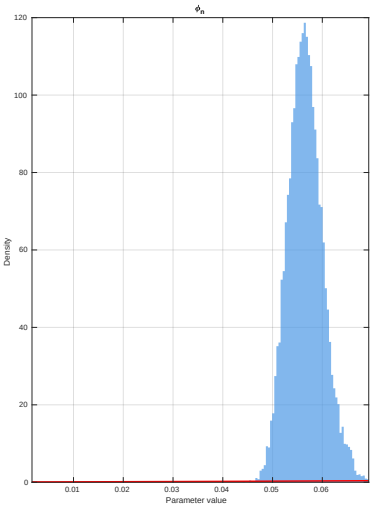

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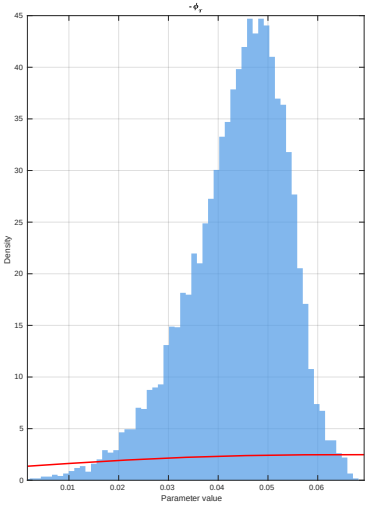

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- MPC (coefficient on net worth n_{t-1}): $\mathbb{E}[1-\beta] = \mathbf{0.056}$
- MPC-DFT (coefficient on $-\tau_t$): $\mathbb{E}[(1-\beta)(1-\theta)] = \mathbf{0.045}$

Point Estimates: Not Very Ricardian!

Specification	Attenuation (θ)	MPC (ϕ_n)	MPC-DFT ($-\phi_\tau$)
Baseline	0.194 [0.019, 0.544]	0.056 [0.051, 0.063]	0.045 [0.025, 0.058]
Non-durable consumption	0.130 [0.013, 0.372]	0.058 [0.052, 0.064]	0.050 [0.036, 0.060]
Survey-based forecasts	0.231 [0.020, 0.657]	0.009 [0.007, 0.011]	0.007 [0.003, 0.010]
Non-separable utility	0.096 [0.029, 0.234]	0.065 [0.050, 0.072]	0.059 [0.039, 0.067]
Variable distortionary taxes	0.597 [0.174, 0.952]	0.058 [0.050, 0.064]	0.023 [0.003, 0.049]
Variable HtM income shares	0.123 [0.012, 0.343]	0.076 [0.069, 0.084]	0.066 [0.050, 0.078]
Including SVAR instruments	0.123 [0.013, 0.262]	0.063 [0.059, 0.067]	0.055 [0.046, 0.062]
Six selected instruments	0.174 [0.017, 0.404]	0.033 [0.028, 0.037]	0.027 [0.019, 0.034]

Notes: Point estimates are medians of the marginal posterior distributions. The [5%, 95%] credible intervals are reported in brackets.

General Equilibrium

GE Model Summary

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  - Capital investment, inelastic labor, risk-free debt
  - Many shocks
  - Cognitive discounting over taxes *only*

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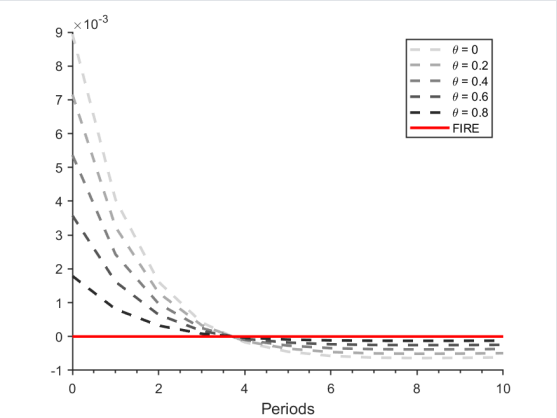

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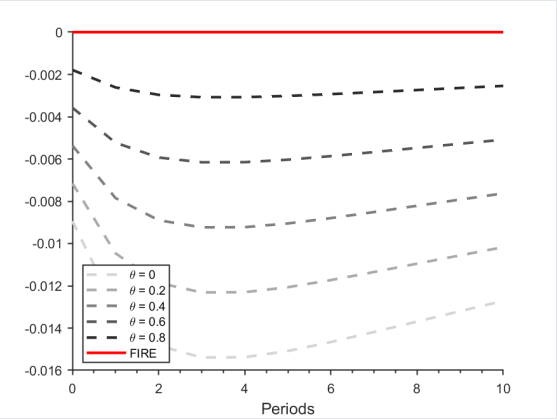
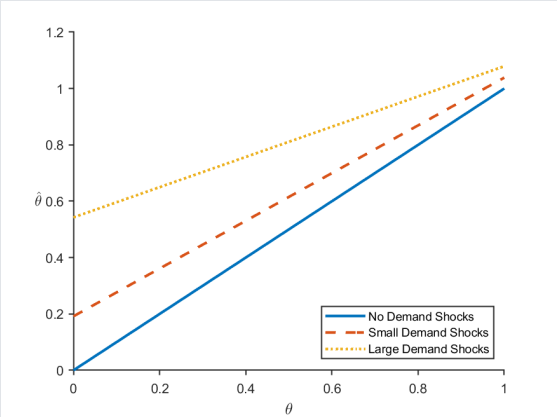
- $\implies$ consumption $\uparrow$
- $\implies$ investment $\downarrow$ $\implies$ recession



Estimates of the Model's Consumption Function



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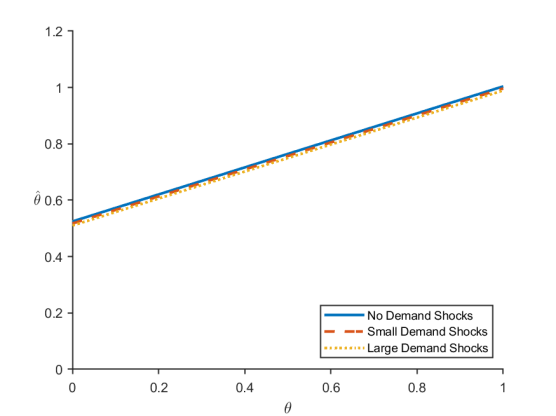

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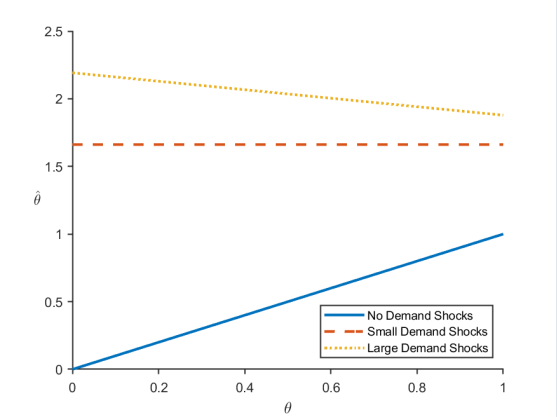

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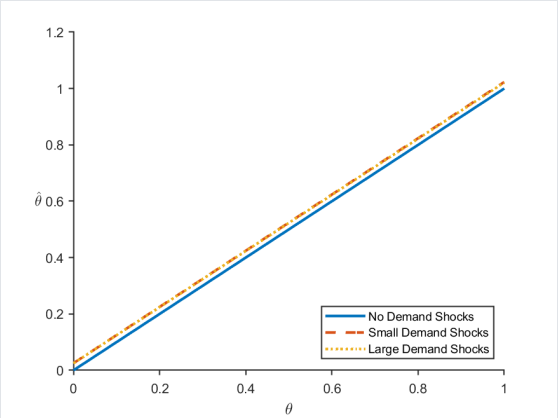

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- Lagged aggregates: only works without demand shocks (or if i.i.d.)
- Macro IVs: always works!

Conclusion

- How Ricardian Are We? *Not very.*
- Behavioral non-Ricardianism is useful, likely, and doesn't require an other-wise non-rational model.
- Serious macro implications!
- **B-HIVE** extremely useful for "identifying modern macro equations with old shocks"
  - We'll release the code (easy to adapt!)
  - Please reuse the library of structural shocks (on my website)

References

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